

Quiz 1 Study Guide

CSC 4780 – Networking & Security Study Notes

1. Networking Fundamentals & Architecture

What Is a Network?

- **Node** → Device (computer, router, phone)
- **Link** → Communication channel (wire, fiber, radio, Bluetooth)
- **Network** = Nodes + Links
- **Internet** = Network of networks

Key Idea

The Internet is a **graph**:

- Vertices → devices
- Edges → connections

Network Architecture

Architecture vs Protocol

- **Architecture** → High-level blueprint
- **Protocol** → Rules for communication
- **Layering** → Break complex problem into manageable layers

Why Layering?

- Abstraction
- Modularity
- Easier debugging
- Replace one layer without breaking others

TCP/IP Model

Layer	Unit	Example
Application	Data	HTTP

Layer	Unit	Example
Transport	Segment	TCP / UDP
Network	Packet	IP
Link	Frame	Ethernet
Physical	Bits	101010

Encapsulation

Each layer wraps the data with its own header.

Link Layer Concepts

Frames

- Bag of bits
- Identified and separated by network adapters
- Used between directly connected hosts

Requirements of a Link

- Framing
- Bit transparency
- Error detection
- Liveness detection
- Address negotiation

Reliable Delivery

Stop-and-Wait Protocol

1. Sender sends frame
2. Waits for ACK
3. Timeout → retransmit

Weakness

- Easily attackable (e.g., delay or drop ACKs)
- Low efficiency

Ethernet

- Shared medium (originally coax)
- Uses **random access**
- Collision detection (CSMA/CD)
- Frame structure includes:
 - Sync
 - MAC addresses
 - Payload

Wireless Issues

Hidden Terminal Problem

A and C cannot hear each other but both send to B → collision at B.

Additional Issues

- Signal fading
- Interference
- Mobility ≠ wireless

2. Internet Protocol (IP)

IP Basics

- Global addressing
- 32-bit IPv4 address
- Interface-based addressing (not host-based)

Structure

Network portion + Host portion

Example:

129.82.138.254

IP Packet Header Fields

- Version
- Header length
- Total length
- TTL (Time To Live)
- Protocol (TCP=6, UDP=17)
- Source & Destination addresses

- Fragmentation fields

Forwarding vs Routing

- **Forwarding** → Local decision (output port)
- **Routing** → Building routing table (global decision)

3. DNS (Domain Name System)

What Is DNS?

- Maps names → IP addresses
- Distributed hierarchical system
- Replaced `hosts.txt`

DNS Hierarchy

Root

└─ TLD (.com, .edu)

└─ domain (tntech.edu)

└─ subdomain (cs.tntech.edu)

DNS Resolution

Two Types

- **Recursive**
- **Iterative**

Components

- Stub resolver (client)
- Recursive resolver
- Authoritative server

Caching

- Responses cached with TTL
- Negative caching supported
- TTL controls lifetime

DNS Vulnerabilities

No Authentication (Original DNS)

DNS Cache Poisoning

- Attacker races legitimate server
- Guesses 16-bit transaction ID
- Sends many forged replies

Birthday Attack

Multiple simultaneous queries increase probability of success.

Kaminsky Attack (2008)

- Query random subdomains
- Start unlimited races
- Eventually poison cache
- Could redirect entire domain

DNSSEC

What DNSSEC Provides

- Authentication
- Integrity
- Chain of trust

What It Does NOT Provide

- Confidentiality
- DDoS protection

DNSSEC Core Components

- **DNSKEY** → Public key
- **RRSIG** → Signature
- **DS** → Delegation Signer (chain of trust)

Chain of Trust

Root → TLD → Domain → Subdomain

4. One-Time Pad (OTP)

Security Goal

Confidentiality against passive attacker.

Properties of OTP

- Key = same length as message
- Key is truly random
- Key used once
- Perfect secrecy

Perfect Secrecy Definition

$$P(M | C) = P(M)$$

Ciphertext gives zero information about plaintext.

Two-Time Pad Attack

If key reused:

$$C1 = M1 \oplus K$$

$$C2 = M2 \oplus K$$

$$C1 \oplus C2 = M1 \oplus M2$$

Attacker can recover plaintext relationships.

OTP Weaknesses

- Key distribution problem
- No authentication
- Malleable (bits can be flipped)

5. Hash Functions

Definition

Maps arbitrary input → fixed-length output.

Example: SHA-256

Desired Properties

- One-way (pre-image resistance)
- Collision resistance
- Target collision resistance

- Pseudo-randomness

Random Oracle Model (ROM)

Theoretical ideal:

- Output is truly random
- Same input → same output
- Otherwise unpredictable

Real hashes approximate ROM.

Applications

- Password storage
- File integrity
- Digital signatures
- Blockchain
- Random number generation

6. BGP – Border Gateway Protocol

Why BGP Exists

Internet scale:

- Millions of prefixes
- Multiple administrative domains
- Need policy control

Autonomous Systems (AS)

- Group of routers under one admin
- Uses:
 - IGP internally
 - BGP externally

BGP Goals

- Reachability
- Loop-free paths
- Policy enforcement
- NOT optimality

Path Vector Protocol

- Sends full AS_PATH
- Loop detection: if own ASN appears → reject

Key BGP Attributes

AS_PATH

- Sequence of AS numbers
- Used for loop prevention and policy

NEXT_HOP

- Where to forward traffic

LOCAL_PREF

- Higher value preferred
- Used within AS

BGP Messages

- Open
- Update
- Keepalive
- Notification

Policies

- Multihomed AS may not want to act as transit
- AS path prepending
- Selective advertisement

7. BGP Attacks

Prefix Hijacking

Announce someone else's prefix.

Subprefix Hijacking

- Announce more specific prefix:

- 10.0.0.0/16 vs 10.0.0.0/24
- More specific wins.

Route Leak

Announcing routes beyond intended scope.

AS Path Manipulation

- Shortening
- Insertion
- Fabrication

8. RPKI (Resource Public Key Infrastructure)

Purpose

Validate origin AS of prefix.

How It Works

1. RIR signs certificate for resource holder
2. Holder creates ROA:
 - Authorized ASN
 - Prefix
 - Max length
3. Validator checks route advertisement.

Validation States

- VALID
- INVALID
- NOT FOUND

Limitations

- No AS_PATH validation
- Doesn't stop route leaks
- Centralized trust model

9. BGPSEC

Goal

Cryptographically validate entire AS_PATH.

Core Idea

Each AS:

1. Signs update before forwarding
2. Signature covers:
 - Prefix
 - Accumulated path
 - Neighbor ASN

Process

At Ingress

- Validate all signatures

At Egress

- Append ASN
- Add signature

Differences from BGP

- Uses `BGPSEC_PATH`
- One NLRI per update
- Signature per AS
- Requires RPKI keys

10. Security Themes Across Course

Infrastructure Trust vs Message Privacy

Mechanism	Protects
DNSSEC	Name integrity
RPKI	Origin validation
BGPSEC	Path validation
PKI	Identity
Encryption	Message confidentiality

Core Takeaways

1. The Internet was built without security.
2. Security often retrofitted.
3. Availability sometimes prioritized over security.
4. Randomness is critical.
5. Key management is harder than algorithms.
6. Perfect security exists in theory (OTP), not practice.
7. Routing security is still evolving.